

module - 1

①

1) Define the following terms with Example: [8m]

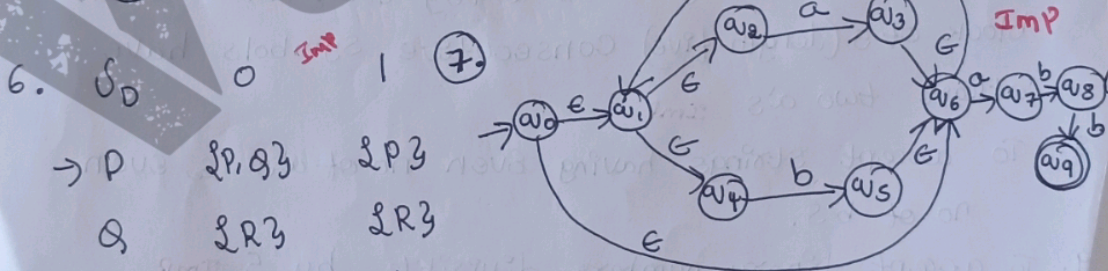
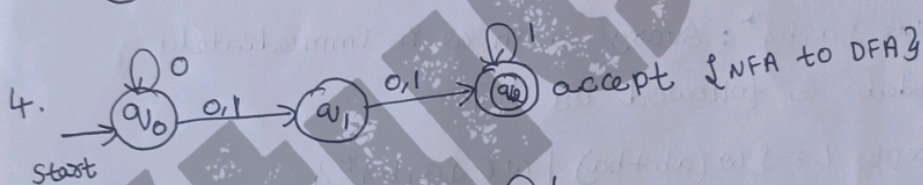
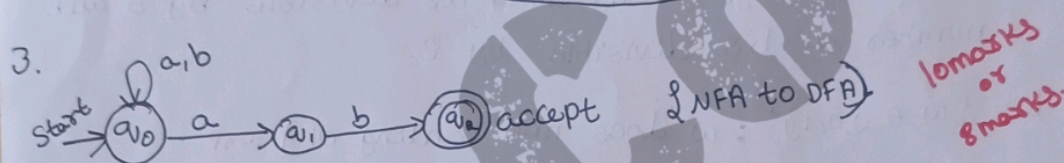
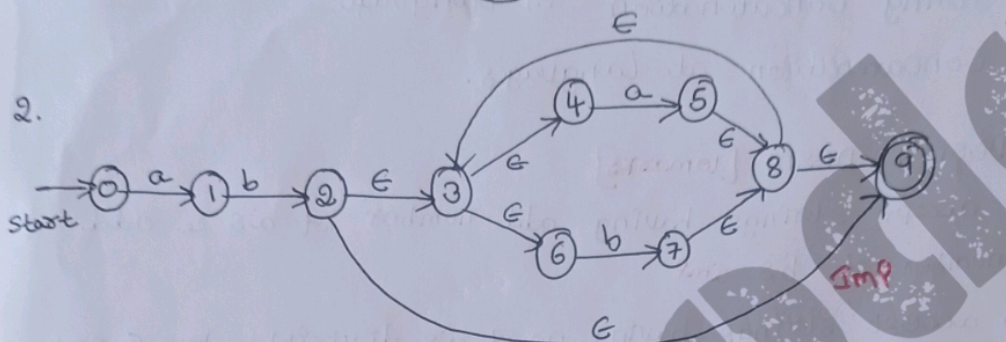
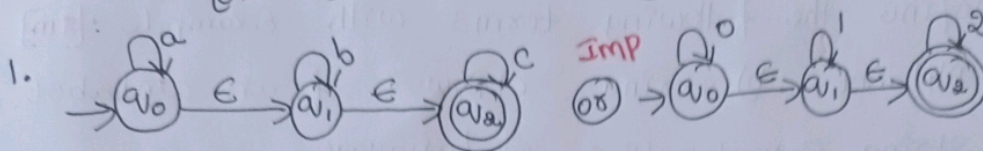
- Alphabet.
- string.
- Power of an alphabet.
- String Concatenation.
- Language.
- Concatenation at Languages.

2) Design DFSM. [10marks]

- To accept strings having odd number of a's & odd number of b's. **Imp**
- To accept strings having no. of a's divisible by 5 and number of b's divisible by 3.
- $L = \{w \in \{0,1\}^* : w \text{ does not end with } 01\}$ **Imp**
- $L = \{w \in \{a,b\}^* : \text{Every } a \text{ in } w \text{ is immediately preceded \& followed by } b\}$
- To accept $L = \{w (ab+ba) \mid w \in \{a,b\}^*\}$
- To accept $L = \{wbab \mid w \in \{a,b\}^*\}$
- To accept strings over $\{a,b\}$ such that each block of 5 (length five) consecutive symbols have at least two a's. **Imp**
- To accept strings having Even no of a's & even no of b's.
- To accept binary numbers divisible by 5 **Imp**
- Obtain a DFA to accept the language
 $L = \{w : |w| \bmod 5 \neq 0\}$ on $\Sigma = \{a,b\}$.

Convert the following ϵ -NDFsm of DFsm

[NFA to DFA conversion]



$\rightarrow P \quad \{P, Q\} \quad \{P\}$
 $Q \quad \{R\} \quad \{R\}$
 $R \quad \{S\} \quad \emptyset$
 $*S \quad \{S\} \quad \{S\} \quad \{NFA to DFA\}$

minimization

Problems [Distinguishable & Indistinguishable]

②

1.

δ	a	b
$\rightarrow A$	B	E
B	C	F
* C	D	H
D	E	H
E	F	I
* F	G	B
G	H	B
H	I	C
* I	A	E

8maxes
or
10maxes

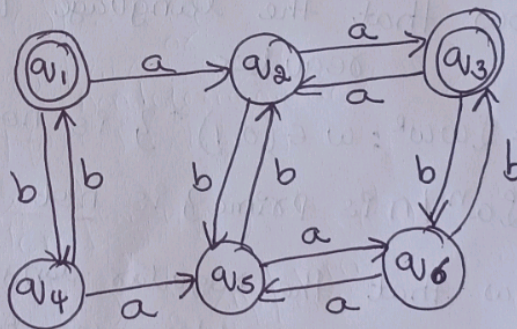
2. Imp

δ	0	1
$\rightarrow A$	B	F
B	G	C
* C	A	C
D	C	G
E	H	F
F	C	G
G	G	F
H	G	C

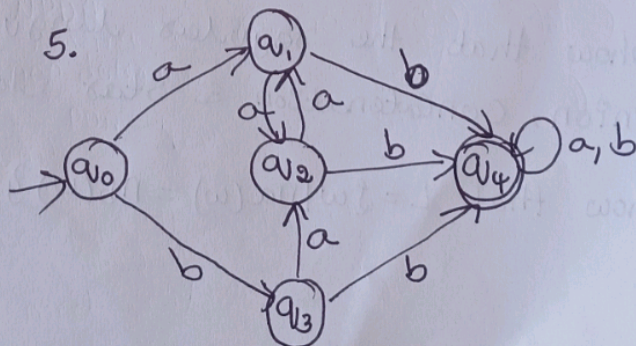
3.

δ	0	1
$\rightarrow A$	B	A
B	A	C
C	D	B
* D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

4. Imp



5.



module - 2

1. State & prove pumping lemma theorem for regular languages. show that [10marks]

$L = \{a^n b^n \mid n \geq 0\}$ is not Regular. Imp

2. Show that regular languages are closed under Complement and intersection. 5m/4m

3. Using Kleen's theorem. prove that for any regular Expression R , there exists a finite automata $M = \{Q, \Sigma, \delta, q_0, F\}$ which accepts $L(R)$. [10marks]

OR

4. Prove Kleen's theorem. Any language that can be defined with a regular Expression can be accepted by some FSM & so is regular.

5. Show that the language $L = \{a^i b^j \mid i > j\}$ is not regular.

6. $L = \{ww^R : w \in (0,1)^*\}$ is not regular. 3m

7. $L = \{a^n \mid n \text{ is prime}\}$ is not regular. 3m

8. Show that the regular languages are closed under homomorphism, difference & complementation 5m

9. Show that the regular languages are closed under union, concatenation & star closure. 5m

10. Show that $L = \{w \mid n_a(w) = n_b(w)\}$ is not regular. 3m

Obtain Regular Expression for the following (3)

languages.

1. $L = \{a^n b^m c^p \mid n \leq 4, m \geq 2, p \leq 2\}$ IMP
2. $L = \{w \mid |w| \bmod 3 = 0 \text{ \& } w \in \{a, b\}^*\}$ (8m)
3. $L = \{a^n b^m \mid m+n \text{ is even}\}$
4. $L = \{w \mid n_a(w) \bmod 3 = 0 \text{ where } w \in (a, b)^*\}$ IMP
5. $L = \{a^{2n} b^{2m} \mid n \geq 0, m \geq 0\}$
6. $L = \{w \mid \text{strings ends with } ab \text{ or } ba \text{ where } w \in \{a, b\}^*\}$
7. $L = \{a^n b^m \mid n \geq 4, m \leq 3\}$
8. $L = \{0^n 1^m \mid m \geq 1, n \geq 1, mn \geq 3\}$
9. $L = \{w \in \{a, b\}^* \mid \text{string with atmost one pair of consecutive a's}\}$ IMP
10. $L = \{a^n b^m \mid m \geq 1, n \geq 1, nm \geq 3\}$

Obtain an NFA for the Regular Expression

1) $a^* + b^* + c^*$ IMP

2) $(a+b)^* aa (a+b)^*$ IMP

3) $(0+1)^* 01 (0+1)^*$

4) $0^* 1^* + (0+1)^* 01$

(8m)
 $\left\{ \begin{array}{l} \text{E-NFA} \\ \text{Const}^n \end{array} \right\}$

module 4

Push Down Automata. [PDA]

1. What is NPDA? Design NPDA for language

$L = \{a^n b^n \mid n \geq 1\}$. Draw transition diagram. write sequence of moves made by NPDA to accept the string aaabbb.

2. Design a PDA for the language. Imp

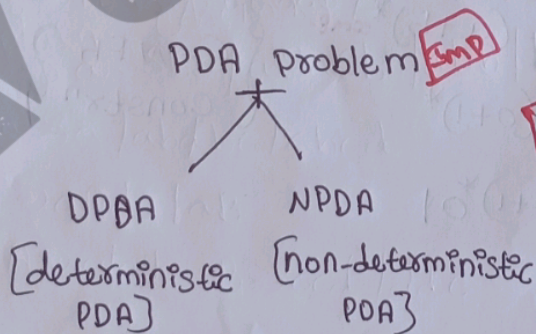
$L = \{w c w^R \mid w \in \{a, b\}^*\}$ where w^R is reverse of w and show the moves made by PDA for the string "aabcbaa" & "abacbbba".

3. Is the PDA to accept the language

$L = \{w w^R \mid w \in (a+b)^*\}$ is deterministic?

4. Is the PDA to accept the language

$L = \{a^n b^{2n} \mid n \geq 1\}$ is deterministic?
or
nondeterministic?



1. Define Context free grammar. Design CFG for the following language.

i) $L = \{0^i 1^j \mid i \neq j \geq 0, j \geq 0\}$ *Imp*

ii) $L = \{a^n b^m \mid n \geq 0, m > n\}$

iii) $L = \{0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0\}$ *Imp*

iv) $L = \{ww^R \mid w \in (a, b)^+\}$

v) $L = \{a^n b^m c^k \mid n + 2m = k \text{ for } m \geq 0 \text{ \& } n \geq 0\}$

vi) $L = \{w : |w| \bmod 3 = 0\}$ over $\Sigma = \{a\}$ *Imp*

vii) $L = \{a^n b^m c^k : m = n + k\}$ over $\Sigma = \{a, b, c\}$

viii) $L = \{w : w \text{ has a substring } ab\}$

2. Define Ambiguity. Consider the grammar *Imp*.

$$E \rightarrow E + E \mid E * E \mid (E) \mid id$$

Find LMD & RMD derivations & parse tree for the string $id + id * id$. Show that the grammar is ambiguous. *10/8 marks*

3. Define Leftmost derivation. Rightmost derivation and Parse tree. Consider the grammar.

$$S \rightarrow AbB \quad A \rightarrow aA \mid \epsilon$$

$$B \rightarrow aB \mid bB \mid \epsilon \quad D \rightarrow a \mid \epsilon$$

Obtain LMD, RMD & Parse tree for the string "aaabab".

4. Define Ambiguity. Consider the grammar
 $E \rightarrow E+E \mid E-E \mid E * E \mid E/E \mid a/b$ Find LMD & RMD
 parse tree for the string $a+b * a+b$. Show that
 the grammar is ambiguous. Imp

5. Is the following grammar ambiguous?

1) $S \rightarrow aB \mid bA$
 $A \rightarrow aS \mid bAA \mid a$
 $B \rightarrow bS \mid aBB \mid b$

2) $S \rightarrow iCtS \mid iCtSeS \mid a$
 $C \rightarrow b$

String "Pbtibtaea"

String "aabbab"

3) $S \rightarrow OS \mid SOS \mid \epsilon$

String "o|o|"

6. Eliminate left recursion from the following grammar

1. $E \rightarrow E+T \mid T$ Imp

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid id$

2) $S \rightarrow Ab \mid a$

$A \rightarrow Ab \mid Sa$

* Define left
 recursion?

7. Explain Chomsky Normal form? (CNF)

8. Define left factoring. Consider the following grammar Imp

1. $S \rightarrow iEtS \mid iEtSeS \mid a$ $E \rightarrow b$

2. $S \rightarrow aSSbs \mid aSaSb \mid abb \mid b$

3. $S \rightarrow a \mid ab \mid abc \mid abcd$

4. $S \rightarrow aAd \mid aB$

$A \rightarrow a \mid ab$ $B \rightarrow ccd \mid ddc$

1. Define Turing machine (TM)? Explain the Turing machine model. *Imp*
2. Design the Turing machine to accept the language $L = \{a^n b^n c^n \mid n \geq 1\}$. Draw the transition diagram & show the moves made by Turing machine for the string "aabbcc". *10/12m*
3. Explain various techniques used for construction of Turing machine. *7m/8m*
4. Explain the following: *Imp*
 - a) multitape Turing machine.
 - b) Non-deterministic TM
 - c) Linear bounded Automata (LBA)
 - d) Post Correspondence problem
 - e) quantum Computers
 - f) Church-Turing Thesis
 - g) Decidable & Undecidable language.
 - h) halting Problem.
 - i) P and NP classes.
 - j) Recursively Enumerable languages.

Short note questions for 20 marks

Design of TM Problems

(6)

1. To accept the language $L = \{0^n 1^n 2^n \mid n \geq 0\}$. Draw the transition diagram. write the sequence of moves made by TM ~~man~~ for a string "001122" Imp [10 marks]
2. $L = \{0^n 1^n \mid n \geq 1\}$ string 0011 & 00111.
3. $L = \{ww^R \mid w \in (a+b)^*\}$